

Snack Analysis Report

1. Project Objective

The objective of this project was to analyze and compare the nutritional content of menu items from McDonald's and Starbucks using PySpark. The analysis focused on determining which brand offers relatively healthier menu options by examining key nutritional attributes such as calories, fat, carbohydrates, fiber, protein, sodium, and sugar.

The project also aimed to demonstrate the use of data engineering techniques such as data cleaning, dataset merging, schema standardization, and aggregation analysis using PySpark.

2. Datasets Used

The project utilized four datasets containing nutritional information about menu items from both brands.

McDonald's Dataset

1.mcd.csv

Contains nutritional information for McDonald's food items such as burgers, breakfast meals, sandwiches, and other menu products.

Starbucks Datasets

Three datasets were provided for Starbucks.

sb_food.csv

sb_drinks.csv

sb_drink_expanded.csv

- sb_food.csv contains nutritional information for Starbucks food products.
- sb_drinks.csv contains beverage nutrition data with a limited set of attributes.
- sb_drink_expanded.csv contains detailed beverage nutrition data including caffeine and vitamin information.

3. Methodology

Before performing the analysis, several preprocessing steps were necessary to prepare the datasets.

Stage 1: Converting Excel Files to CSV

The original datasets were provided in Excel format. Since PySpark works more efficiently with CSV files, all Excel files were converted into CSV format.

This ensured:

- Faster loading in PySpark
- Simpler schema inference
- Better compatibility with distributed processing

Stage 2: Dataset Profiling

Prior to preprocessing, all four datasets were profiled to evaluate their structure, identify potential data quality issues, and determine the cleaning requirements. The profiling process included duplicate record detection and missing-value assessment.

The analysis revealed that the **mcd.csv**, **sb_food.csv**, and **sb_drink_expanded.csv** datasets did not contain any duplicate records. However, **22 duplicate records** were identified within the **sb_drinks.csv** dataset. Missing-value analysis further identified a single missing caffeine value in the Starbucks Expanded Drinks dataset. Additionally, several caffeine entries were recorded as "**Varies**" and "**varies**", while placeholder values represented by "-" were present within the Starbucks Nutrition Drinks dataset.

These findings provided the basis for the subsequent cleaning and standardization procedures performed in the next stage.

Stage 3: Data Cleaning and Standardization

Based on the issues identified during profiling, several preprocessing operations were performed to improve data quality and consistency. The column "**Unnamed: 0**" was renamed "**Item_Name**" to improve readability and standardization. Placeholder values represented by "-" were converted to **NULL** values, and nutritional attributes were converted to appropriate numeric data types to support statistical analysis.

The **22 duplicate records** identified in the **sb_drinks.csv** dataset were removed to prevent duplicate counting during aggregation. Special handling was also applied to caffeine-related information, where values recorded as "**Varies**" and "**varies**" were preserved as valid entries rather than being incorrectly treated as missing values.

Stage 4: Common Schema Creation and Dataset Integration

The four datasets contained different column names, structures, and nutritional attributes. To enable direct comparison, a unified schema was created across all datasets.

To enable a fair comparison, the columns were standardized across all datasets to a common schema containing:

Brand
Source_File
Category
Item_Name
Calories, Fat
Carbohydrates
Fiber
Sugars
Protein
Sodium and Caffeine.

After standardization, all four datasets were merged into a single integrated dataset named `master_nutrition_df`, enabling direct comparison across products from both brands.

Stage 5. Exploratory Data Analysis (EDA)

After the completion of data cleaning, standardization, and schema integration, Exploratory Data Analysis (EDA) was performed on the consolidated dataset to better understand the nutritional characteristics of menu items from McDonald's and Starbucks.

The final integrated dataset consisted of 770 records, including 260 McDonald's products and 510 Starbucks products. This unified dataset provided a consistent foundation for comparative nutritional analysis across both brands.

The EDA phase focused on examining the overall composition of the dataset, identifying data quality patterns, validating the success of schema integration, and generating descriptive nutritional statistics. Average values for key nutritional attributes such as calories, fat, carbohydrates, fiber, sugars, protein, and sodium were calculated and analyzed across different datasets and product categories.

A detailed missing-value assessment was also conducted to evaluate data completeness. This analysis revealed that certain nutritional attributes were unavailable in specific source files due to differences in the original datasets. Rather than replacing missing values with artificial estimates, all unavailable values were preserved as `NULL` to maintain data integrity and prevent misleading analytical results.

The exploratory analysis further confirmed that duplicate records identified during the profiling stage had been successfully removed and that all datasets were correctly transformed into the common schema. Special attention was given to caffeine-related information, where values recorded as "**Varies**" were preserved as valid categorical entries instead of being incorrectly treated as missing values.

The integrated dataset contained a total of 770 records and served as the final analytical dataset used for all subsequent nutritional comparisons and aggregations performed in this study.

6. Nutritional Comparison Results

The following table shows the average nutritional values calculated for menu items from both brands.

To ensure a fair comparison, food products and beverage products were analyzed separately. Comparing beverages directly with food products could produce misleading conclusions because the nutritional composition and serving purposes of these product categories differ substantially.

Product Type Counts

Brand	Product Type	Count
McDonalds	Drink	150
McDonalds	Food	110
Starbucks	Drink	397
Starbucks	Food	113

Drinks Comparison

Brand	Items	Avg Calories	Avg Fat	Avg Carbs	Avg Sugars	Avg Sodium
McDonalds	150	299.47	7.73	50.27	44.64	128.43
Starbucks	397	180.38	2.85	104.05	32.96	20.00

The beverage analysis showed notable differences between the two brands. Starbucks beverages contained approximately 40% fewer calories and significantly lower sodium levels compared to McDonald's beverages. Although Starbucks drinks contained higher average carbohydrate content, they generally exhibited lower sugar levels and lower fat content. These findings suggest that Starbucks beverages may be more suitable for consumers prioritizing calorie reduction and sodium control.

Food Comparison

Brand	Items	Avg Calories	Avg Fat	Avg Carbs	Avg Protein	Avg Sodium
McDonalds	110	462.09	22.94	43.36	20.78	996.64
Starbucks	113	356.64	16.35	41.49	11.47	N/A

The food comparison revealed a different nutritional pattern. Starbucks food products contain fewer calories and lower fat levels on average. However, McDonald's food products provide substantially higher protein content, which may be advantageous for consumers seeking higher-protein meal options. Fiber content was relatively similar between the two brands, suggesting that neither brand demonstrated a strong advantage in this category.

Overall Brand Comparison

Brand	Items	Avg Calories	Avg Fat	Avg Sugars	Avg Protein
McDonalds	260	368.27	14.17	29.42	13.34
Starbucks	510	226.70	6.40	32.96	7.77

7. Limitations

The analysis was performed using the nutritional attributes available in the source datasets. Certain nutritional metrics were unavailable in some Starbucks datasets, particularly sodium and sugar information for Starbucks food products. As a result, comparisons involving these attributes were limited to the available data.

Additionally, the study focused on average nutritional values and did not consider serving size differences, ingredient quality, or broader dietary factors that may influence overall healthfulness.

8. Conclusion

The analysis demonstrated that Starbucks products generally contain lower calories, fat, and sodium levels, while McDonald's products provide higher protein content. Separate evaluations of food and beverage products revealed that nutritional differences vary considerably across product categories.

Starbucks beverages exhibited lower average calories, fat, sugar, and sodium levels, whereas McDonald's food products provided substantially higher protein content. These findings highlight the importance of evaluating multiple nutritional attributes rather than relying on a single metric when assessing healthfulness.

Overall, the results suggest that neither brand can be considered universally healthier across all nutritional dimensions. The relative healthiness of a menu item depends on the nutritional goals and dietary priorities of the consumer.